# Rishab Vempati

925-854-8943 | rishabvempati.2004@gmail.com | linkedin.com/in/rvempat | rishabvempati.com

## Education

**Purdue University** — West Lafayette, IN
*B.S. in Computer Science & Artificial Intelligence, GPA: 3.83/4.00* — *Aug. 2023 – May 2027*

## Experience

**Software Engineering Intern** — Aug. 2025 – Present
*Ekai (Startup)* — *Remote*

- Built Microsoft Teams integration for enterprise AI assistant, adapting existing Slack implementation to support Teams-specific features including automated task execution and knowledge base querying.
- Integrated Microsoft Graph API with RAG pipeline to enable document retrieval across 10,000+ files using Pinecone vector database, achieving 92% retrieval accuracy.
- Optimized semantic search with hybrid SQL retrieval and caching layer, reducing P95 response latency from 8.2s to 1.8s for 50+ concurrent users.

**Software Engineering Intern** — Jun. 2025 – Aug. 2025
*AssetMark* — *San Francisco, CA*

- Designed and implemented migration of legacy Enterprise Communication Service from monolithic .NET/C# to event-driven microservices using Azure Service Bus, processing 1M+ emails quarterly with 99.7% delivery success rate.
- Designed fault-tolerant message handling with dead-letter queues and exponential backoff retry policies, reducing message failure rate from 0.8% to 0.12% (85% reduction) and eliminating manual intervention for errors.
- Decoupled 5 service dependencies into independently deployable microservices with API gateways, enabling horizontal autoscaling that handled traffic spikes during quarter-end reporting periods.

**Software Engineering Intern** — May 2024 – Aug. 2024
*Accelera Payments (Startup)* — *San Francisco, CA*

- Implemented Kafka streaming pipeline for ISO 20022 payment processing with partition-based load balancing and exactly-once delivery semantics, load-tested to 50K+ daily transactions.
- Containerized 4 microservices using Docker with multi-stage builds in GitLab CI/CD pipelines, reducing deployment time from 2 hours to 25 minutes and enabling zero-downtime blue-green releases.
- Built integration test suite and Grafana dashboards monitoring message throughput (avg 850 msg/sec), processing latency (P99: 120ms), and error rates across distributed system.

**Undergraduate Researcher – Data Science** — Jan. 2024 – May 2024
*Cisco* — *West Lafayette, IN*

- Constructed demand forecasting models using ensemble methods (Random Forest, XGBoost) with feature engineering on 2M+ records, improving MAPE by 18% over baseline ARIMA models across 10,000 product SKUs.
- Optimized ETL pipeline using Apache Spark for distributed processing of supply chain datasets, implementing incremental loading that reduced execution time from 45 minutes to 18 minutes.

## Projects

**Parking Lot Detection & Mapping** | *Python, PyTorch, OpenCV, Docker, AWS* | *GitHub*

- Collaborated on 5-person team developing U-Net CNN in PyTorch for satellite image segmentation, achieving 84% IoU accuracy on parking lot detection across 10K+ validation images.
- Built automated pipeline ingesting 50GB+ Bing Maps imagery; containerized training workflow with Docker, reducing training time by 60%; contributed 5,000+ annotations to OpenStreetMap.

**YourPingMe – Intelligent Notification Scheduler** | *Python, React, Redis, Celery, PostgreSQL* | *GitHub*

- Designed task automation platform using Redis and Celery for scheduling recurring LLM API calls with cron expressions; built notification system delivering real-time SMS/email alerts upon task completion.

## Technical Skills

**Languages**: Python, Java, C++, C, C#, SQL, JavaScript, TypeScript, Bash, R, HTML/CSS, Assembly
**Technologies**: FastAPI, Flask, Node.js, .NET, Docker, Kubernetes, Apache Kafka, Redis, CI/CD, Linux, Azure, AWS (EC2, S3, Lambda), PostgreSQL, MongoDB
**AI/ML**: PyTorch, TensorFlow, JAX, OpenCV, Pinecone, LangChain, Elasticsearch, LangGraph, MCP, Spark
**Certifications**: AWS Certified Cloud Practitioner, Microsoft Azure Fundamentals (AZ-900)